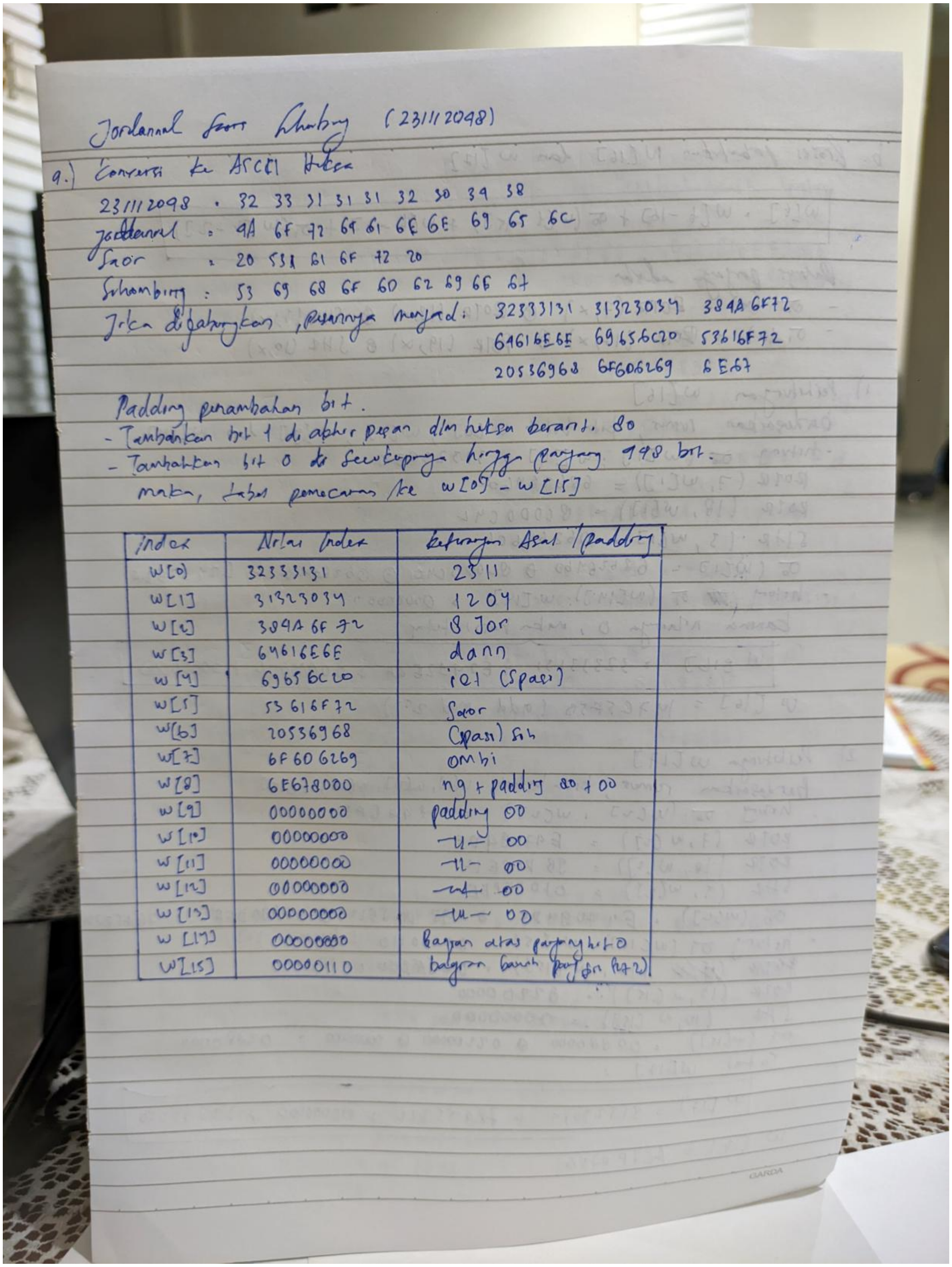
Jordanniel Saor Sihombing (231112048)

A. Tuliskan tahapan mengisi W[0] sd. W[15]



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a.) Konversi ke ASCII Heksa

231112048 : 32 33 31 31 31 32 30 34 38

Jordanniel : 4A 6F 72 64 61 6E 6E 69 65 6C

Saor : 20 53 61 6F 72 20

Sihombing : 53 69 68 6F 6D 62 69 6E 67

Jika digabungkan, pesannya menjadi: 32333131 31323034 384A6F72 64616E6E 69656C20 53616F72 20536968 6F6D6269 6E67

Padding penambahan bit.

- Tambahkan bit 1 di akhir pesan dlm heksa berarti 80
- Tambahkan bit 0 di selanjutnya hingga panjang 448 bit.

maka, tabel pemecahan ke w[0] - w[15]

index	Nilai Index	Keterangan Asal / padding
w[0]	32333131	2311
w[1]	31323034	1204
w[2]	384A6F72	8 Jor
w[3]	64616E6E	dann
w[4]	69656C20	iel (spasi)
w[5]	53616F72	Saor
w[6]	20536968	(spasi) Sih
w[7]	6F6D6269	ombi
w[8]	6E678000	ng + padding 80 + 00
w[9]	00000000	padding 00
w[10]	00000000	-11- 00
w[11]	00000000	-11- 00
w[12]	00000000	-11- 00
w[13]	00000000	-11- 00
w[14]	00000000	bagian atas panjang bit 0
w[15]	00000110	bagian bawah panjang bit (272)

B. Tunjukkan proses pembentukan W[16] dan W[17]

b. Proses pembentukan W[16] dan W[17]

$$\boxed{W[t] = W[t-16] + \sigma_0(W[t-15]) + W[t-7] + \sigma_1(W[t-2])}$$

Rumus penjangnya adalah

- $\sigma_0(x) = ROTR(7, x) \oplus ROTR(18, x) \oplus SHR(3, x)$
- $\sigma_1(x) = ROTR(17, x) \oplus ROTR(19, x) \oplus SHR(10, x)$

1) Perhitungan W[16]

berdasarkan rumus, kita butuh W[0], W[1], W[9] dan W[14]

- hitung $\sigma_0(W[1])$; W[1] = 31323034

  ROTR(7, W[1]) = 68626460

  ROTR(18, W[1]) = 8C0D0C4C

  SHR(3, W[1]) = 06264606

  $\sigma_0(W[1])$ = 68626460 ⊕ 8C0D0C4C ⊕ 06264606 = E2492E2A

- hitung ~~⊠~~ $\sigma_1(W[14])$: W[14] = 00000000

  karena nilainya 0, maka hasil tetap

$$\boxed{W[16] = 32333131 + E2492E2A + 00000000 + 00000000}$$

W[16] = 147C5F5B (add mod 2^{32})

2) Perhitungan W[17]

Berdasarkan rumus, kita butuh W[1], W[2], W[10], W[15]

- hitung $\sigma_0(W[2])$, W[2] = 384A6F72

  ROTR(7, W[2]) = E4709 4DE

  ROTR(18, W[2]) = 9B DC8E12

  SHR(3, W[2]) = 07094DEE

  $\sigma_0(W[2])$ = E4709 4DE ⊕ 9BDC8E12 ⊕ 07094DEE = 78A55722

- hitung $\sigma_1(W[15])$: W[15] = 00000110 (desimal 272)

  ROTR ~~(17,~~ (17, W[15]) = ~~00880000~~ 00880000

  ROTR(19, W[15]) = 02200000

  SHR(10, W[15]) = 00000000

  $\sigma_1(W[15])$ = 00880000 ⊕ 02200000 ⊕ 00000000 = 02A80000

  Total W[17] :

$$\boxed{W[17] = 31323034 + 78A55722 + 00000000 + 02A80000}$$

W[17] = AC7F8756

C. Tunjukkan iterasi perhitungan variabel kerja A, B, … ,G, H untuk iterasi i = 0 sd 2 (3 iterasi)

C. Iterasi Perhitungan Variabel Kerja A sd H (i = 0, 1, 2)

SHA-256 menggunakan 8 variabel kerja dgn "Initial Hash Value"

- Inisialisasi awal (i = 0)

~~A//////~~ A = 6A09E667, B = BB67AE85, C = 3C6EF372,

D = A54FF53A, E = 510E527F, F = 9B05688C, G = 1F83D9AB,

H = 5BE0CD19

Iterasi i = 0

(menggunakan W[0] = 32333131, K[0] = 428A2F98)

1. Hitung $\Sigma_1(E)$:

ROTR(6,E) $\oplus$ ROTR(11,E) $\oplus$ ROTR(25,E)

= FD443993 $\oplus$ 50821C44 $\oplus$ 887293DD = 25B4B63D

2. Hitung CH = $(E \wedge F) \oplus (\neg E \wedge G)$

= (510E527F $\wedge$ 9B05688C) $\oplus$ (AEF1AD80 $\wedge$ 1F83D9AB)

= 1104400C $\oplus$ 0E818980 = 1F85C98C

3. Hitung temp1 = H + Σ_1(E) + CH + K[0] + W[0]

= 5BE0CD19 + 25B4B63D + 1F85C98C + 428A2F98 + 32333131

= 15D8ADAB

4. Hitung $\Sigma_0(A)$: ROTR(2,A) $\oplus$ ROTR(13,A) $\oplus$ ROTR(22,A)

: 9A027999 $\oplus$ 3385 04F3 $\oplus$ 99DA0279 : 706D7F13

5. Hitung MAJ : $(A \wedge B) \oplus (A \wedge C) \oplus (B \wedge C)$

= 2A01A605 $\oplus$ 2808E262 $\oplus$ 3866A200 = 3A6FE667

6. Hitung Temp2 : Σ_0(A) + MAJ

= 706D7F13 + 3A6FE667 = AADD657A

Update Variabel (Hasil akhir i = 0) :

H = G = 1F83D9AB

G = F = 9B05688C

F = E = 510E527F

E = D + temp1 = A54FF53A + 15D8ADAB = BB28A2E5

D = C = 3C6EF372

C = B = BB67AE85

B = A = 6A09E667

A = temp1 + temp2 = 15D8ADAB + AADD657A = C0B61325

Iterasi i = 1

menggunakan W[1] = 31323034, K[1] = 71374491

1. Σ_1(E) = 16C5C2BB $\oplus$ 5CBB20A2 $\oplus$ B655BB28 = FE9E7101

2. CH = 11080265 $\oplus$ 00057808 = 110D4A6D

3. temp1 = 1F83D9AB + FE9E7101 + 110D4A6D + 71374491 + 31323034

= D1993985

4. $\Sigma_0(A)$ = 502D84C9 $\oplus$ 264BB16C $\oplus$ 84C9502D = F2AFA588

5. MAJ = 40000C85 $\oplus$ 80260205 $\oplus$ 2A01AG05 = EA27A25

6. temp 2 = F2AFA588 + EA27A425 = DCD749AD

update variabel hasil akhir i = 1

H = 9B05688C

G = 510E527F

F = BB28A2E5

E = 3C6EF372 + D1990DE = 0F07FD50

D = BB67AE85

C = 6A09E667

B = C0B61325

A = D19989D5 + DCD749AD = AE70538B

Iterasi i = 2

(menggunakan W[2] = 3896F72, K[2] = B5C0FBCF)

1.) $\Sigma_1(E)$ = 40381FF5 $\oplus$ FA11C0FF $\oplus$ A01C0FFA = 1A35D0F0

2) CH = 0A00A040 $\oplus$ 5108026F = 5B08A26F

3) Temp 1 = 9B05688C + 1A35D0F0 + 5B08A26F + B5C0FBCF + 38946F72

= FE4F472C

4) $\Sigma_0(A)$ = EB9C14E2 $\oplus$ A7175CE0 $\oplus$ CF0A7175 = 82813977

5) MAJ = 80301501 $\oplus$ 2A0092D5 $\oplus$ 40000225 = EA305527

6) temp 2 = 82813977 + EA305327 = 6CB18C9E

update variabel (hasil akhir i = 2)

H = 510E527F

G = BB28A2E5

F = 0F07FD50

E = BB67AE85 + FE4F472C = B9B6F5B1

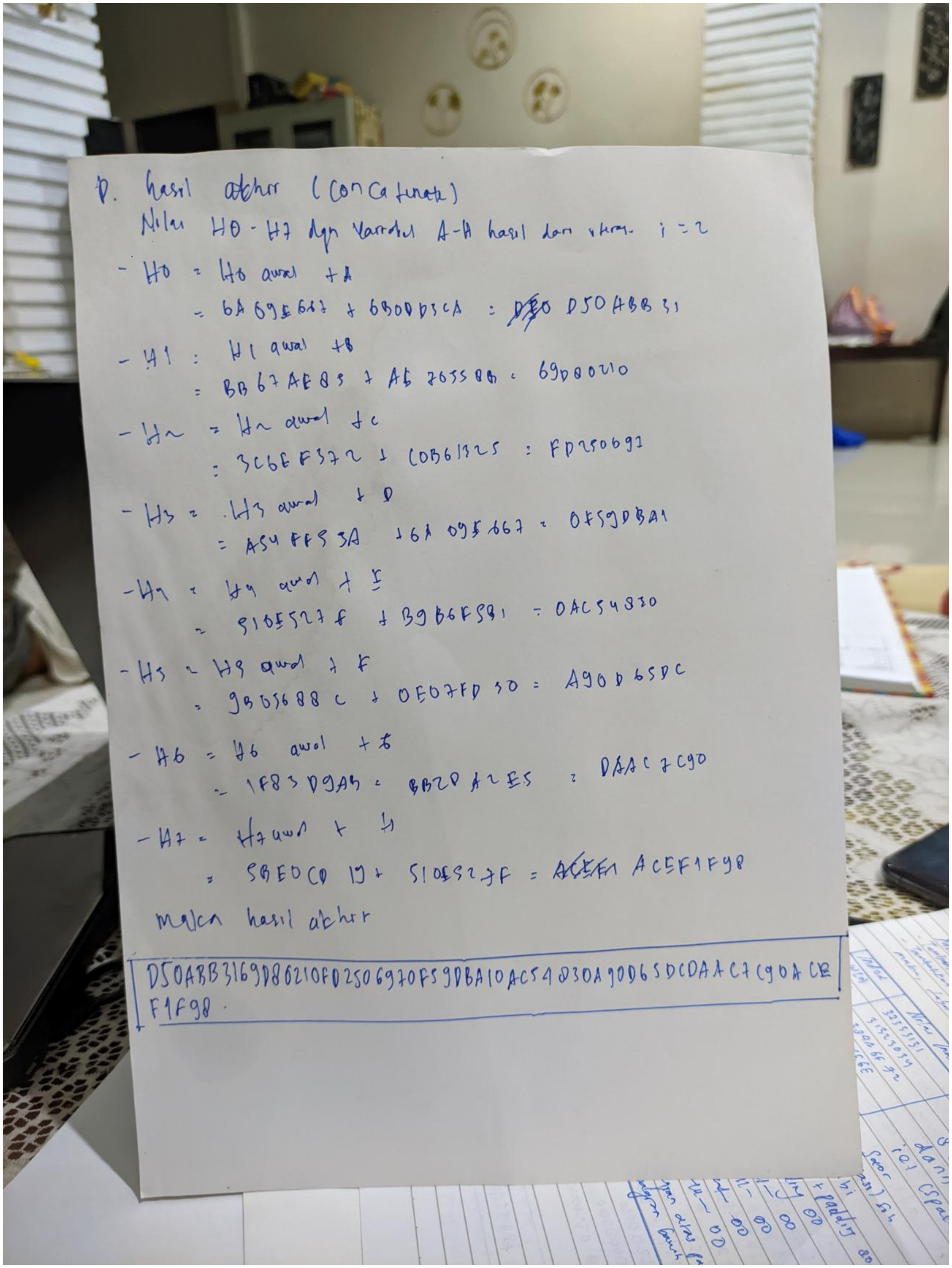
D = 6A09E667

C = C0B61325

B = AE70538B

A = FE4F472C + 6CB18C9E = 6B00D5CA

D. Hasil = H0|H1|….|H6|H7 (***concatenate***)



D. hasil akhir (concatenate)

Nilai H0 - H7 dgn variabel A-H hasil dari iterasi i = 2

- H0 = H0 awal + A

= 6A09E667 + 6B0DD5CA = ~~D50~~ D50ABB31

- H1 : H1 awal + B

= BB67AE85 + AE7055B6 = 69D80210

- H2 = H2 awal + C

= 3C6EF372 + C0B61325 = FD250697

- H3 = H3 awal + D

= A54FF53A + 6A09E667 = 0F59DBA1

- H4 = H4 awal + E

= 510E527F + B9B6F581 = 0AC54830

- H5 = H5 awal + F

= 9B05688C + 0E07FD30 = A90D65DC

- H6 = H6 awal + G

= 1F83D9AB + BB20A2E5 = DAA47C90

- H7 = H7 awal + H

= 5BE0CD19 + 510E527F = ~~ACEF1~~ ACEF1F98

maka hasil akhir

$\boxed{\text{D50ABB3169D80210FD2506970F59DBA10AC54830A90D65DCDAA47C90ACEF1F98.}}$